

Tools of the Trade: Linux and SQL

Activity: Add and Manage Users with Linux Commands

Introduction

In this activity, I practiced managing users and groups in Linux using the Bash shell.

I learned how to add a new user, assign the user to a primary group, change file ownership, add the user to another group, and finally remove the user from the system.

The main commands I used were `useradd`, `usermod`, `chown`, `userdel`, and `groupdel`.

Activity Scenario

In this lab, a new employee with the username `researcher9` joined an organization.

The task was to manage their access during their time in the organization.

I first added the user to the system and assigned them to the `research_team` group. I then made the user the owner of a project file.

Later, the employee's role changed, so I added them to the `sales_team` group as a secondary group.

Finally, the employee left the organization, so I removed their account and the unused group associated with the account.

Lab Environment

- **Course:** Google Cybersecurity Certificate
 - **Course:** Tools of the Trade: Linux and SQL
 - **Activity:** Add and Manage Users with Linux Commands
 - **Operating System:** Linux
 - **Shell:** Bash
 - **User:** `analyst`
-

Task 1: Add a New User

The first task was to add the new employee to the Linux system.

I used:

```
sudo useradd researcher9
```

This created the user account `researcher9`.

Next, I assigned the user to the `research_team` group as their primary group.

I used:

```
sudo usermod -g research_team researcher9
```

The `-g` option is used to set the user's primary group.

At this point, `researcher9` was added to the system and assigned to the Research group.

Task 2: Assign File Ownership

The employee needed to take responsibility for a project file.

The file was located at:

```
/home/researcher2/projects/project_r.txt
```

I changed the owner of the file to `researcher9` using:

```
sudo chown researcher9 /home/researcher2/projects/project_r.txt
```

The `chown` command is used to change the ownership of a file or directory.

This allowed `researcher9` to become the owner of the project file.

Task 3: Add the User to a Secondary Group

Later, `researcher9` started working with both the Research and Sales departments.

Their primary group remained `research_team`, but I needed to add them to `sales_team` as a secondary group.

I used:

```
sudo usermod -a -G sales_team researcher9
```

Here, `-a` means to append the group without removing the user's existing group memberships, while `-G` specifies the supplementary group.

After this change, `researcher9` was a member of both the Research and Sales groups.

Task 4: Delete the User

After the employee left the organization, their account needed to be removed from the system.

I used:

```
sudo userdel researcher9
```

The command displayed a message indicating that the `researcher9` group was not removed because it was not the user's primary group.

This was expected in the lab.

I then removed the unused `researcher9` group with:

```
sudo groupdel researcher9
```

This cleaned up the group that was no longer needed.

Commands I Practiced

Command	What I used it for
<code>useradd</code>	Add a new user
<code>usermod</code>	Modify a user's group membership
<code>chown</code>	Change file ownership
<code>userdel</code>	Delete a user
<code>groupdel</code>	Delete an unused group
<code>sudo</code>	Run commands with administrative privileges

What I Learned

This activity helped me understand how users and groups are managed in Linux.

I practiced creating a user account and assigning a primary group. I also learned how to change ownership of a file and add a user to a secondary group.

Finally, I practiced removing a user and cleaning up an unused group.

I also learned that Linux user and group management commands generally require administrative privileges, which is why `sudo` was used throughout the lab.

Cybersecurity Connection

User and group management is an important part of access control in cybersecurity.

Organizations need to make sure that users only have the access they need for their roles. When someone joins an organization, their account and permissions need to be configured. If their role changes, their group membership may also need to change.

When an employee leaves an organization, their account should be removed so they can no longer access company systems.

The commands used in this activity are basic Linux tools that can help administrators manage authentication and access to system resources.

Demonstration

The screenshots below show my practical work during the lab, including creating `researcher9`, assigning the primary and secondary groups, changing file ownership, and removing the user and unused group.

```
analyst@9633d59385f2:~$ sudo useradd researcher9
analyst@9633d59385f2:~$ sudo usermod -g research_team researcher9
analyst@9633d59385f2:~$ sudo useradd researcher9 -g research_team
useradd: user 'researcher9' already exists
analyst@9633d59385f2:~$ sudo chown researcher9 /home/researcher2/projects/project_r.txt
analyst@9633d59385f2:~$ sudo usermod -a -G sales_team researcher9
analyst@9633d59385f2:~$ sudo userdel researcher9
userdel: group researcher9 not removed because it is not the primary group of user researcher9.
analyst@9633d59385f2:~$ sudo groupdel researcher9
analyst@9633d59385f2:~$
```

Conclusion

In this activity, I practiced basic Linux user and group management.

I learned how to add users, assign groups, change file ownership, add secondary group access, and remove users from the system.

These are important Linux fundamentals for cybersecurity because managing user accounts and access is a key part of protecting systems and resources.

Course Information

Google Cybersecurity Certificate

Course: Tools of the Trade: Linux and SQL

Activity: Add and Manage Users with Linux Commands

Disclaimer

This document is part of my personal cybersecurity learning portfolio. The activity was completed for educational purposes as part of the Google Cybersecurity Certificate.